Keypad Door Lock System

An Arduino-based electronic access control project

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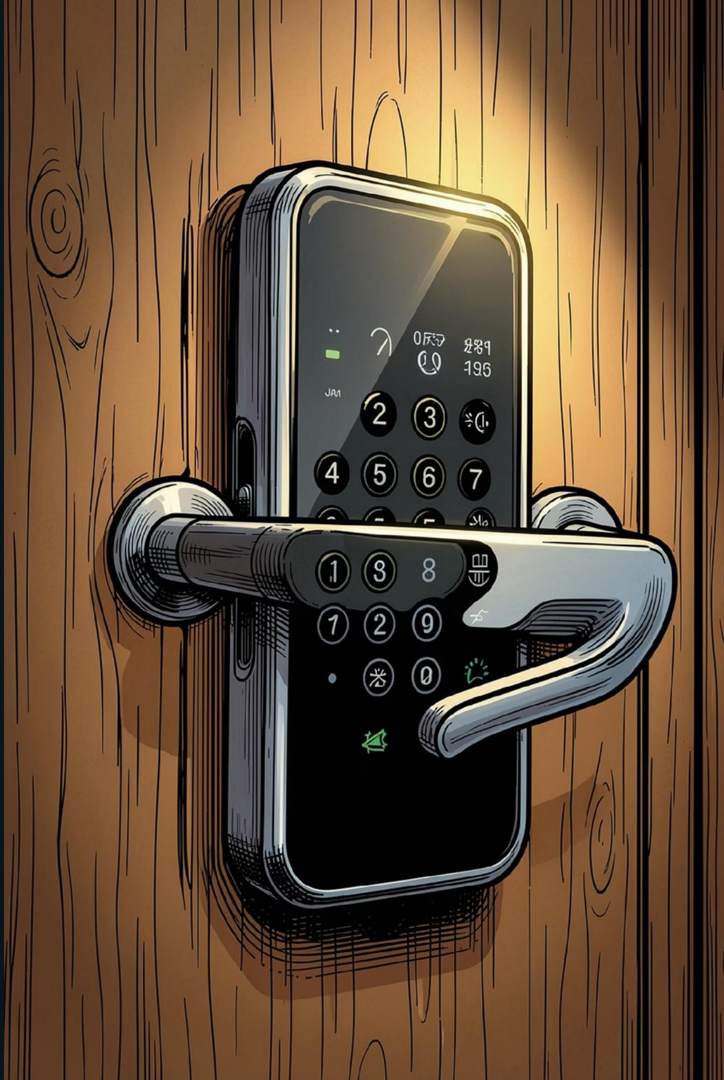


The Security Challenge

The Problem

Traditional locks are vulnerable to picking, bumping, and lost keys — a constant risk for homes and offices.

The Need

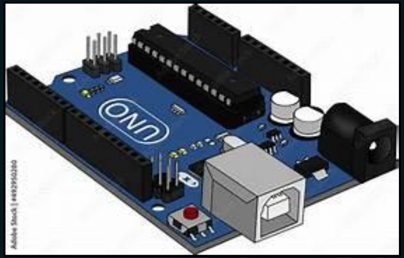
Digital security solutions are essential in today's world. Password-protected systems offer a smarter, safer alternative.

The Goal

Build a reliable, low-cost, password-protected electronic lock using accessible Arduino components.

System Components

Four key components work together to create a complete access control system.



Arduino Uno

The central brain — processes input, runs logic, and controls all outputs.

4×4 Keypad

Input interface for entering the passcode — 16 keys for digits and commands.

16×2 LCD

Visual feedback display showing prompts, access status, and error messages.

Servo Motor

Mechanical actuator that physically locks and unlocks the door latch.

Existing System vs. Proposed Solution

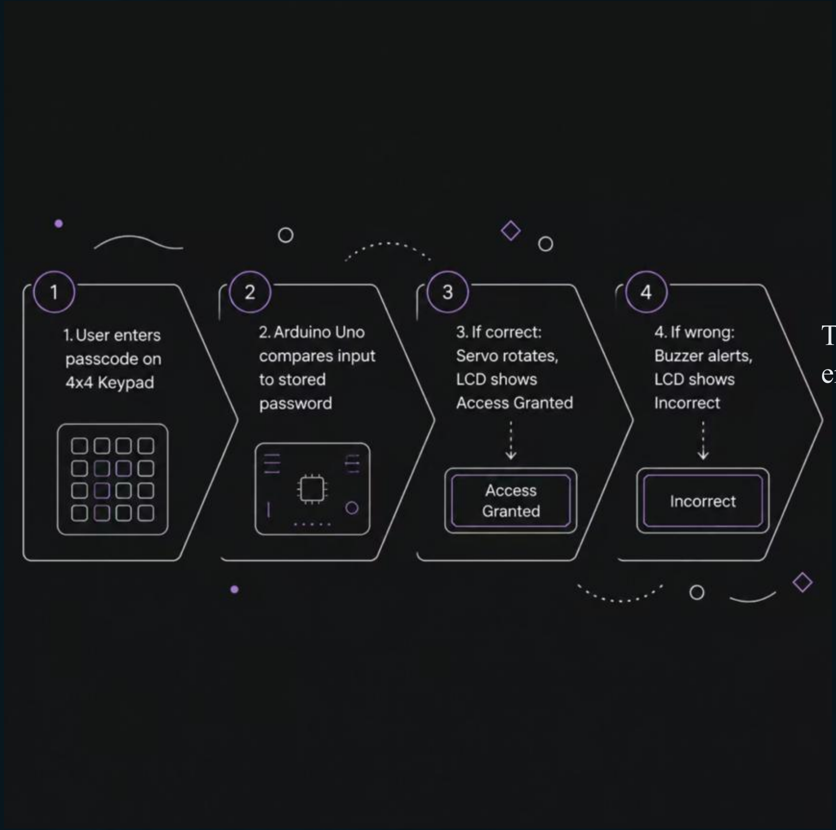
Existing System

- Mechanical key-and-lock mechanism
- Keys can be copied or lost easily
- No audit trail or access logging
- Fixed access — cannot be updated remotely
- Vulnerable to lock-picking and bumping

Proposed Solution

- Microcontroller-driven electronic keypad lock
- Programmable password — no physical key needed
- LCD display confirms access status in real time
- Password can be changed by authorized users
- Servo motor ensures reliable, automated locking

How It Works

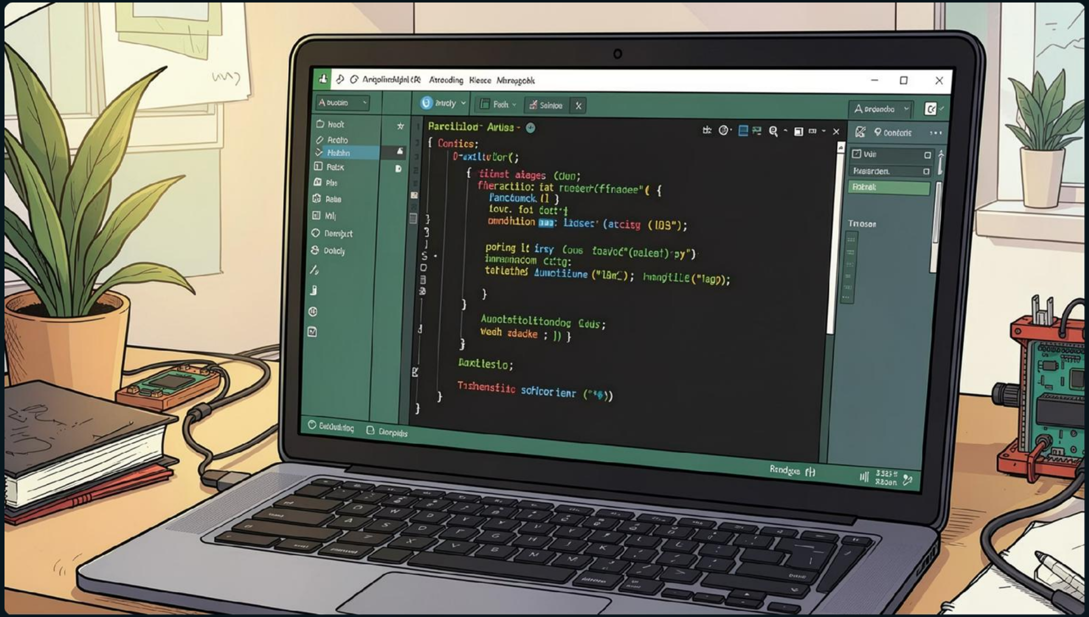


Step-by-Step Logic

The system continuously monitors the keypad. When a 4-digit code is entered, the Arduino validates it in real time.

- **Correct code:** Servo rotates to unlock; LCD confirms access
- **Wrong code:** Buzzer triggers an alert; LCD displays "Incorrect"
- System resets automatically after a short delay

Implementation & Coding



Core Libraries Used

A **4-digit master password** is hardcoded. The program handles both successful authentication and failed attempts with appropriate feedback.

Keypad.h

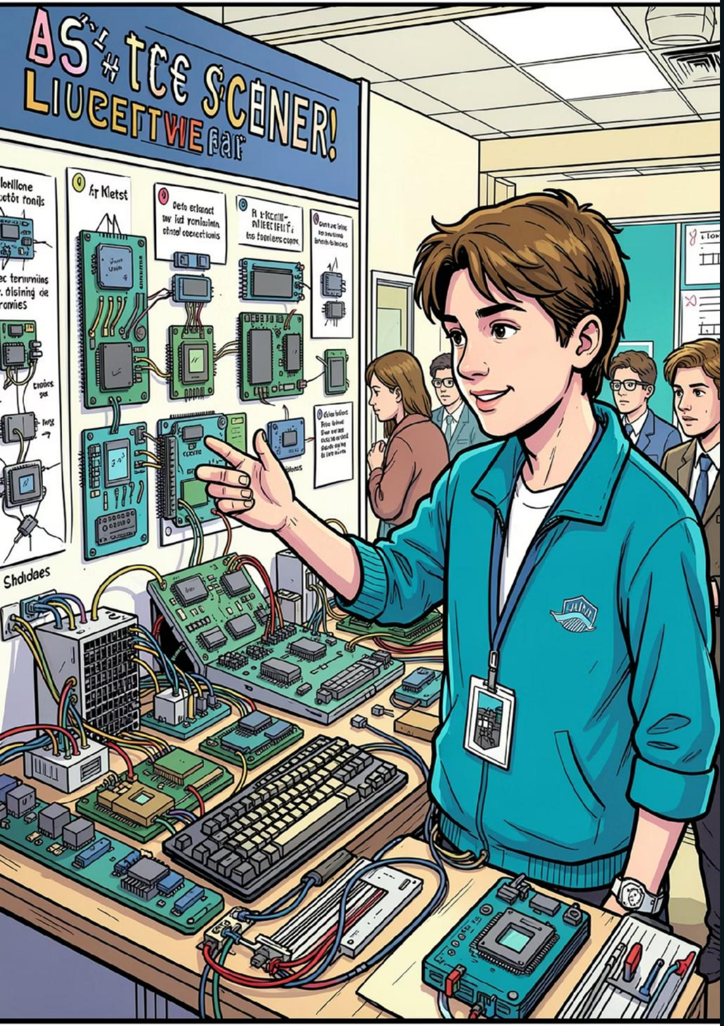
Reads key presses from the 4×4 matrix

LiquidCrystal.h

Controls the 16×2 LCD display output

Servo.h

Drives the servo motor for locking action



Project Outcome

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Components

Integrated seamlessly into one system

100%

Digital Auth

No physical keys required

The system successfully demonstrated automated access control — proving that embedded electronics can deliver real-world security solutions at a fraction of commercial cost.

Conclusion & Future Scope

What We Proved

This project shows how DIY electronics can effectively replace traditional manual locks with a smart, programmable alternative.

- Low-cost, customizable, and reliable
- Scalable to real-world applications
- Foundation for smart home systems

Future Upgrades

→ **Fingerprint Sensor**

Biometric authentication for higher security

→ **Facial Recognition**

AI-based identity verification module

→ **IoT Connectivity**

Remote unlock via smartphone app

Thank You !

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